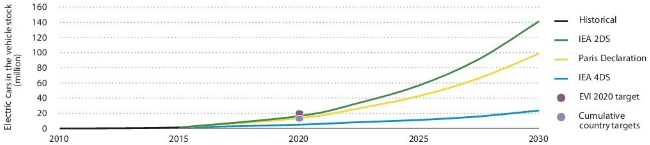
The stock and deployment of electric cars by 2030: A projection



Individual country commitments would bring 13 million electric cars on the road by 2020. The EVI aims at a deployment of 20 million electric cars by 2020. In both cases, reaching 2020 deployment targets for Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) requires a sizeable growth of the electric car stock. Meeting 2030 decarbonisation and sustainability goals requires a major deployment of electric cars in the 2020s.

Source: Global EV Outlook 2016

an ambitious plan launched by the Ministry of Heavy Industries, which is set to change the way transportation is perceived in India. It aims to get seven million electric and hybrid vehicles plying on Indian roads by 2020. To drive this change, the government is offering incentives to new manufacturing units interested in the technology; R&D support is being provided for players who want to get into segments like EV technology, power electronics, motors, systems integration, testing infrastructure; and help is provided to the EV industry to participate and represent India at various trade shows. Nationally, the government is aiming to set up charging infrastructure with the help of local electric supply units, and is encouraging retrofits wherever possible.

The battery for EVs, the largest component in the vehicle, currently accounts for nearly 50 per cent of the overall cost, which is substantially lower than what it was six years ago, and prices are dropping even further. One area of concern is that China, Argentina, Australia and Chile have the highest lithium reserves across the world, while India doesn't have any viable lithium deposits. So, if we need to grow Indian businesses and make them competitive, much more research needs to be done to find lithium alternatives. One potential option is the aluminium battery, which Stanford University has reported can reduce charging time and mitigate the various health hazards associated with lithium-ion batteries.

In short, while India is at the cusp of growth in the EV segment, it would need to introspect and invest in newer technologies.

What to expect in the future

With new players coming in, backed with adequate funding, we should see some cost-effective, integrated and innovative solutions. Knowledge-sharing will be critical in speeding up technology development and adoption. Collaborative commerce will be a building block for the industry, and it will be championed by government initiatives in sustainable transportation.

For the global automobile segment, the forecast is a CAGR of more than 7 per cent over the next two years.. This figure, backed by the right government initiatives, should make the EV segment attractive to any Indian entrepreneur.

Send us your feedback at: editsec@efy.in

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References: SIAM, Global Outlook on E – Vehicles report – 2016, Government Policy Declaration sheets and Documents